SWIFT SAFE AND SUSTAINABLE

GURUGRAM METRO RAIL LIMITED (GMRL)

GURUGRAM METRO PROJECT

DESIGN BASIS REPORT FOR STATION

Version-R0

August 2025

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TABLE OF CONTENTS

SWIFT SAFE AND SUSTAINABLE

1. INTRODUCTION

1.1. Brief Presentation of Project

This proposed elevated corridor, comprising a 26.65 km main corridor and a 1.85 km spur, connects HUDA Millennium City Centre and Cyber City, with an additional link from Basai Village to Dwarka Expressway. This project, which includes a total of 27 stations and a depot connection, will revolutionize transportation in Gurugram. DDC scope of work includes providing detailed design consultancy services for Civil, Architectural, and E&M works for the entire 28.5 km elevated section.

S. No.	Description	Details
1	Elevated Station	27 No.
2	Corridor Length	28.5 kM
3	Gauge	Standard 1435mm
4	Traction System	750V DC Current

In station portion, the Detailed Design work includes design of Twenty Seven elevated stations named mentioned in below table for MCC-Cyber City Corridor(28.5 KM) of Gurugram Metro Project.

Table 1: List of station

SL. No.	Station Name
1	Millennium City Centre
2	Sector 45
3	Cyber Park
4	District Shopping Center, Sector 47
5	Subhash Chowk
6	Sector 48
7	Sector 33
8	Hero Honda Chowk
9	Udyog Vihar Ph 6
10	Sector 10
11	Sector 37
12	Basai Village

13	Sector 9
14	Sector 7
15	Sector 4
16	Sector 5
17	Ashok vihar
18	Sector 3
19	Bajghera Road
20	Palam Vihar Extension
21	Palam Vihar
22	Sector 23A
23	Sector 22
24	Udyog Vihar Phase-IV/Pandav Chowk
25	Udyog Vihar Phase-V/Phase-I
26	Cyber City/Moulsari Avenue/Belwedre
27	Sector 101

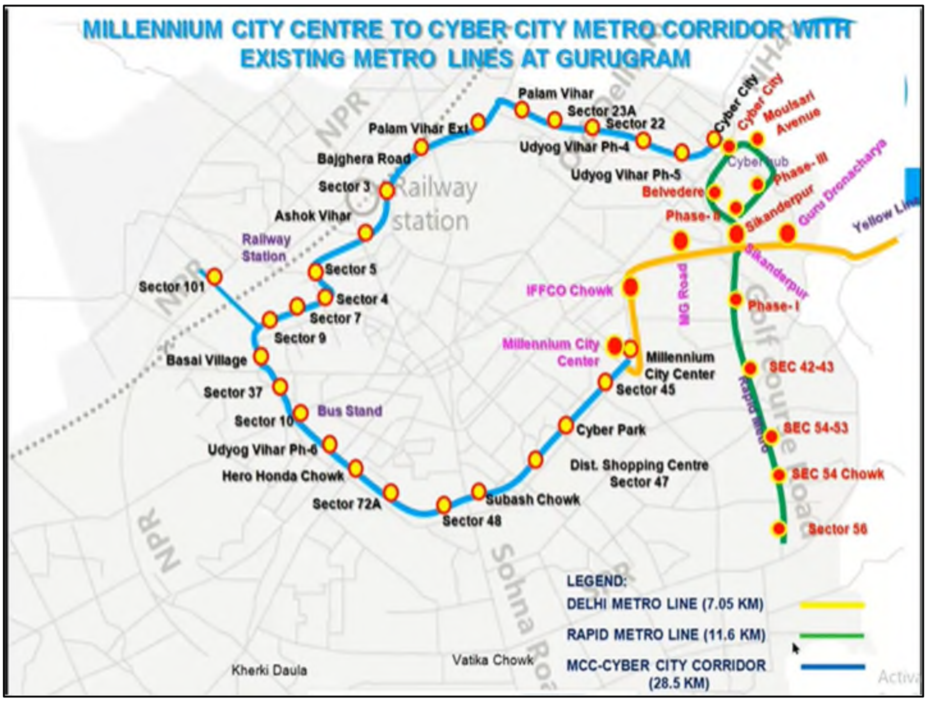

Figure 1: Proposed Corridor of MCC-CYBER CITY(GURUGRAM)

1.2 Scope

The object of this Design Basis Report is to establish a common procedure for the design of "Elevated Metro Stations of GMRL". This is meant to serve as a guide to the designer but compliance with the rules there-in does not relieve them in any way of their responsibility for the stability and soundness of the structure designed. The design of Elevated Metro Stations requires an extensive and thorough knowledge and entrusted to only to specially qualified engineers with adequate practical experience in structure designs.

The structures supporting Metro Live load shall be designed in line with Design Basis Report for Viaduct. Design Basis Report for viaduct shall be followed for following structures/elements of station:

- For single pier and portal pier stations
- Concourse & platform PSC pier arm for single pier stations
- Structural elements which support metro live loads

This design basis report is applicable for following structures/structural elements:

- Structural elements of station which do not support metro live loads.
- Ancillary and entry exit structures separated from main station by an expansion joint.

The DL, SIDL & LL considered in design of station has been specified in 2.4 of this document.

Prestressed concrete structures shall be designed as per IS: 1343. RCC Structures shall be designed by IS: 456. Steel structure design shall be designed by IS: 800. Seismic design shall generally be governed by IS: 1893-Part 1.

1.3 Units

The main units used for design shall be: [t], [m], [mm], [kN], [KN/m2], [MPa], [°C], [rad]

1.4 CODES

All relevant codes as listed in DBR shall be of latest revision including all amendments & corrections.

The design & analysis of structures/members, without Metro live loads, will be governed by following codal requirement:

- Plain and Reinforced Concrete structures: IS 456
- Prestressed concrete structures: IS 1343, IRS CBC
- Steel structures: IS 800- 2007, IS 4923-1997, IS 2062-2011, IS 1161, IS 11384, IS 3935
- Seismic design shall be governed by IS:1893 (Part-I), IS:4326
- Wind loads shall be taken as per IS 875 (Part II and V), IS 875 (Part III)
- Ductile detailing shall be done as per IS 13920
- Design And Construction of Foundations in Soils : IS 1904, IS 2911 (Part 1/Section 1)
- Determination Of Bearing Capacity of Shallow Foundations : IS 6403
- Settlement of Shallow Foundation : IS 8009 (Part I)
- Settlement of Deep Foundation : IS 8009 (Part II)
- High Strength Deformed Steel Bars And Wires For Concrete reinforcement Specification : IS 1786

- Water Retaining Structures : IS 3370 (Part I), IS 3370 (Part II)

2. DESIGN SPECIFICATION FOR STATION BUILDING

2.1 Materials

2.1.1 Cement

For plain and reinforced concrete structures cement shall be used as per clause 5.1 of IS:456 and in case of pre-stressed concrete structures as per clause 5.1 of IS: 1343.

2.1.2 Concrete

As per clause 6, 7, 8, 9 and 10 of IS:456 in case of Plain and Reinforced Concrete structures and Clause 6, 7, 8, 9 and 10 of IS: 1343 for Pre-stressed concrete Structures.

Short term modulus of elasticity (Ec) shall be taken as per cl. 6.2.3.1 of IS:456 for Plain and Reinforced Concrete structures and IS: 1343 for Pre-stressed concrete structures.

The modular ratio for concrete grades shall be taken as per Annex B of IS:456. The Density of concrete shall be as per IS:456.

2.1.3 Prestressing Steel for Tendons

As per clause 5.6.1 of IS: 1343.

2.1.3.1 Young's Modulus

As per Cl. 5.6 of IS: 1343.

2.1.3.2 Prestressing Units

As per clause 13 of IS: 1343.

2.1.3.3 Maximum Initial Prestress

As per clause 19.5.1 of IS: 1343.

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2.1.3.4 Sheathing

As per clause 12.2 of IS: 1343.

2.1.3.5 Density

Weight of strands shall be as per relevant clauses of IS codes as per material being used as indicated in para 2.1.3 above

2.1.4 Structural Steel

Structural steel used shall confirm to

a) Hollow steel sections as per IS:4923

b) Steel for General Structural Purposes as per IS: 2062

c) Steel tubes for structural purpose shall be as per IS:1161

Note:

(i) Grade of steel to be used shall be indicated, shall not be less than minimum grade as applicable, based on whether structure is taking moving loads or not and relevant code as indicated in note (ii) and (iii) below.

(ii) Design of steel structure will be governed by IRS Steel Bridge Code in case structure is taking moving loads of Metro, otherwise will be governed by IS:800. In case of composite (steel-concrete) structures it will be governed by IS:11384 & IS:3935.

(iii) Fabrication shall be done in accordance with IRS B1 (Fabrication Code) in case structure is taking moving loads of Metro, otherwise shall be done as per IS: 800.

2.1.5 Reinforcement

As per clause 5.6 of IS: 456 for Plain and Reinforced concrete structures and as per clause 5.6.2 of IS: 1343 for Pre-stressed concrete structures.

Note: For Seismic zone III, IV & V HYSD steel bars having minimum elongation of 14.5 percent and conforming to requirements of IS:1786 shall be used.

2.1.5.1 Reinforcement Detailing

All reinforcement shall be detailed in accordance with clause 12 and 26 of IS:456 & SP-34 for Plain and Reinforced concrete structures and as per Clause 12.3 and 19.6.3 of IS:1343 and SP-34 for prestressed concrete structures. Ductile detailing of seismic resisting RC elements, shall comply with ductile requirements of IS:13920.

2.2 Durability

Durability of Concrete shall be as per clause 8.0 of IS: 456 for Plain and Reinforced Concrete structures, as per clause 8.0 of IS:1343 for prestressed, Concrete structures and Section 15 of IS:800 for Steel Structures.

2.2.1 Concrete Grades

Minimum grade of concrete for structural elements shall be as follows:

Sl. No.	Structural Components	Minimum Grade of Concrete (cube)
1.	RCC Track Beams	M40
2.	Slab at track level	M40
3.	Beam at Concourse, PD & Platform level	M35
4.	Slab at Concourse, PD & Platform level	M35
5.	Columns	M35
6.	Pile & Pile Cap	M35
7.	Open Foundations	M35
8.	Retaining Walls/Water Tank walls	M35
9.	Crash barrier	M40
10	Prestress Member	M55

2.2.2 Cover to Reinforcement

The cover to reinforcement shall be provided as per clause 26.4 of IS:456 for Plain and Reinforced Concrete Structures and clause 12.3.2 of IS:1343 for prestressed concrete structures or as per clause 15.9.2 of IRS:CBC (Latest – 2014), whichever is applicable as per structural compatibility. Cover to prestressing steel shall be in accordance with clause 12.1.6 of IS: 1343 -2012. Minimum clear cover correspond to fire resistance shall be as per clause 26.4.3 and table 16A of IS 456-2000.

2.2.3 Fire Resistance period

All the structural elements in the station building shall be designed for a minimum fire resistance period of 2 hours. The minimum element thickness for this fire resistance shall be as per clause 21 of IS:456 for Concrete structures and as per Section 16 of IS:800 for Steel structures.

2.2.4 Crack Width Check

All structural concrete elements shall be designed to prevent excessive cracking due to flexure, early age thermal and shrinkage. Flexural crack width shall be checked in accordance with clause 35.3.2 and 43 of IS:456 for Plain and Reinforced Concrete Structures and clause 20.3.2 and 24.2 of IS: 1343 for Prestressed Concrete structures.

2.3 Clearances

i) **Clearance for Road Traffic:** As per relevant IRC specifications and Road Authority requirements. As per relevant IRC , Minimum Vertical Clearance of 5.50m at 0.25m (0.225m (width of crash barrier) + 0.025m (clearance between crash barrier and pier shaft)) from pier shaft outer line.

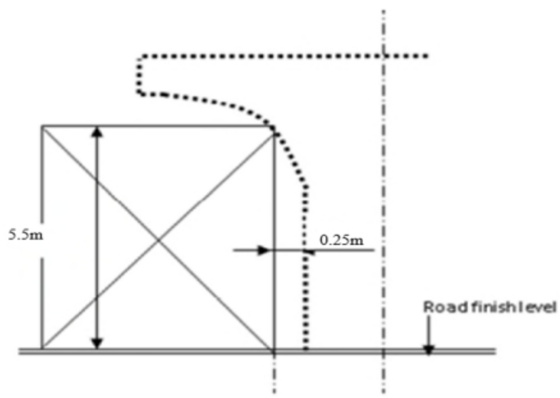

Figure 2: Diagram for vertical clearances

ii) **Clearance for Railway Traffic:** Indian Railway Schedule of Dimensions (SOD) shall be applicable.

iii) **Clearance for Metro Traffic:** As per approved SOD of Gurugram Metro Rail Project.

iv) **For utility services:** The clearances to utilities, drainage etc shall be as mandated by the utility owner/ department.

SWIFT SAFE AND SUSTAINABLE

2.4 Design Loads

Following Elementary loads to be considered according to respective codal provision for design:

Dead Loads	DL
Super Imposed Loads	SIDL
Imposed (Crowd Live) Loads	LL
Earthquake Loads	EQ
Wind Loads	WL
Collision/Impact Loads/Derailment Loads	CL*
Construction & Erection Loads	EL
Temperature Loads	OT
Shrinkage	S
Creep	C
Earth & Water Pressure	EP
Surcharge Loads (Traffic, building etc.)	SR
Pre-stress Force	PR
Long Welded Rail Force	LWR
Differential Settlement	DS

***Load as applicable shall be taken.**

2.4.1 Dead Loads

Self-weight of structures is computed using the unit weights of material mentioned in Section 2.1 of this report. Detailed design will be carried out on the basis of actual framing plan and loading finalized during the design process.

2.4.2 Superimposed Dead Loads (SIDL)

Superimposed dead loads include all the weights of materials on the structure that are not structural elements but are permanent.

<u>For platform slab, the following assumptions will be taken:</u>

- Floor finishes : 3.6 kN/m2 uniform load (density 24 kN/m3)
- Suspension load : 2.0 kN/m2 uniform loads (Suspension load will be considered as load of false ceiling, plumbing & electrical equipment's, Escalator Pits etc. This load is applicable wherever necessary)
- Light partition wall load : 1.0 kN/m2 uniform load
- PSD (Platform Screen Door): As per contractor's specifications
- Platform roof load: 0.75 KN/m2 (As per table 2 of IS 875 (Part II))

<u>For the concourse area, the following assumption will be taken:</u>

- Floor finishes: 3.6 kN/m2 uniform loads.
- Load due to additional fill in the toilets (Light weight Concrete with maximum density of 10 KN/m^3)
- Lift and Escalator support shall be designed as per manufacturer's details.
- Wall load will be applied (200 mm thick Block Wall of Maximum Density 10 KN/m^3)as per the actual & 20 mm plaster on both side of the wall. (Wall load shall be applied, referring the wall location as per the architecture drawings).

- UPS room – 25 KN/m2
- ASS room – 15 KN/m2
- Other Technical rooms – 10 KN/m2
- Glazing load will be applied as per the actual after referring the Architecture drawings.
- Loads to be considered for installation of Solar panel & skylights to be provided in station roof
    - Live load = 75 kg/m2 with reduction as per roof slope.
    - Standing seam roof system with insulation = 25 to 30 kg/m2
    - Solar panel = 30kg/m2

NOTE

- The wall loads will be taken based on actual location shown in architectural drawings. External wall load/glazing load will be taken as per details provided in architectural drawings. It is proposed to take 200mm thick Block Wall of Maximum Density 10 KN/m^3 with 20mm thick plaster on either side. However, same shall not be taken less than 2.4 kN/m2.
- Above assumptions are indicative only. Actual loading calculations to be followed while designing

2.4.3 Imposed (Crowd Live) Load

Imposed loads-on station buildings are those arising from occupancy and the values includes, normal use by persons, furniture and moveable objects, vehicles, rare events such as concentrations of people and furniture, or the moving or stacking of objects during times of re-organization and refurbishment, shall be as per clause 19.3 of IS 456.

2.4.4 Earthquake Loads

Earthquake design shall follow the seismic requirements of IS:1893 (Part-I). The provision as per Design Basis Report for Viaduct of Metro System shall be followed where structures are taking moving loads of metro.

2.4.4.1 Drift Limitation

The storey drift in the building shall satisfy the drift limitation specified in cl. 7 .11.1 in IS:1893 (Part-I).

2.4.4.2 Seismic Detailing

i) For reinforced concrete structures, as per IS:13920.
ii) For other structures, as per IS:4326.

2.4.5 Wind Loads

The wind load shall be calculated as per IS:875 (part III).

2.4.6 Collision/Impact Loads/Derailment Loads

(i) For Road traffic as per IRC 6.
(ii) For Metro as per IRS Bridge Rule.

2.4.7 Construction and erection Loads

The weight of all temporary and permanent materials together with all other forces and effects which can operate on any part of structure during erection shall be taken into account. Allowances shall be made in the permanent design for any locked in stresses caused in any member during erection.

2.4.8 Temperature

As per Cl. 19.5 of IS: 456. Temperature gradient shall be considered as per CL. 215 of IRC-6, if applicable.

2.4.9 Shrinkage

The shrinkage strains shall be evaluated as per clause 6.2.4 of IS:456 for plain and reinforced concrete structures and clause 6.2.4 of IS:1343 for prestressed concrete structures.

For structures supporting Metro loading the effects of shrinkage as per Cl. 5.2.3 of IRS CBC shall be considered.

2.4.10 Creep

The creep strains shall be evaluated as per clause 6.2.5 of IS :456 for plain and Reinforced Concrete structures and clause 6.2.5 of IS :1343 for prestressed concrete structures.

For structures supporting Metro loading the effects of creep as per Cl. 5.2.4 of IRS CBC shall be considered.

2.4.11 Earth & Water Pressure

In the design of structures or part of structure below ground level, such as retaining walls and underground pump room/water tank etc. the pressure exerted by soil or water or both shall be duly accounted for. When a portion or whole of the soil is below the free water surface, the lateral earth pressure shall be evaluated for weight of soil diminished by buoyancy and the full hydrostatic pressure (As per IS:875 (Part V)). All foundation slabs/footings/subjected to water pressure shall be designed to resist a uniformly distributed uplift equal to the full hydrostatic pressure. Checking of overturning of foundation under submerged condition shall be done considering buoyant weight of foundation.

If any of the structure supporting Metro loading is subjected to earth pressure, the loads and effects shall be calculated in accordance with Cl. 5.7 of IRS-Substructure code.

2.4.12 Surcharge Load

In the design of structures or parts of structures below ground level, such as retaining walls and underground pump room/water tank etc. the pressure exerted by surcharge from stationary or moving load, shall be duly accounted for.

2.4.13 Pre-stressing Force (PR)

The prestressing force should be as per IS 1343-2012 for non-track supporting structures. For track supporting structure as per IRS-CBC & GMRL Viaduct DBR.

2.4.14 Long welded Rail Force

Guidelines vide BS Report No. 119 “RDSO Guidelines for carrying out Rail- Structure interaction studies on Metro system (version-2) shall be followed.

A rail structure interaction [RSI] analysis is required because the continuously welded running rails are continuous over the deck expansion joints. The interaction occurs because the rails are directly connected to the decks by fastening system.

2.4.15 Settlement

Maximum and differential settlement shall not exceed, as provided in Table 1 of IS:1904.

2.4.16 Other Forces and Effects

As per clause 19.6 of IS:456.

2.5 Design Load Combinations

2.5.1 Ultimate Load Combinations

Each component of the structure shall be designed and checked for all possible combinations of applied loads and forces. They shall resist effect of the worst combination. Following shall be considered:

(i) Load combinations and factors as per Table 18 of IS:456 for Plain and Reinforced Concrete Structures.
(ii) Load combination and factors as per Table 7 of IS:l343 for prestressed concrete structures.
(iii) Load combination as per Section 3 and factors as per Section 5 of IS:800 for Steel structures.
(iv) Load combination as per clause 6.3 of IS:1893 (Part-I).
(v) Load combinations as per IRS CBC and RDSO guidelines for Seismic design of Railway Bridges where Metro live loads are applicable.

Note:(i) Load combination for construction load case shall be decided by Metro as per methodology of construction.
(ii) Reference of IRC:6 to be taken for collision case if collision of road vehicles are involved .

2.5.2 Serviceability Load Combinations

The following load combinations and load factors shall be used for design for serviceability limit state:
(i) Load combinations and factors as per Table 18 of IS:456 for Plain and Reinforced Concrete Structures.
(ii) Load combination and factors as per Table 7 of IS:l343 for prestressed concrete Structures.
(iii) Load combination as per Section 3 and factors as per Section 5 of IS:800 for Steel structures.
(iv) Load combinations as per IRS CBC where Metro live loads are applicable.

2.6 Deflection Criteria

The deflection limitations as per clause 23.2 of IS:456 for Plain and Reinforced Concrete Structures and Clause 20.3.1 of IS:1343 for Prestressed concrete structures for non-track supporting structures and as per IRS CBC for track supporting structures shall be followed.

For PES / PEB elements deflection limits of members shall generally be as per relevant clause of IS: 800. Maximum deflection limit for purlin is to be considered as Span/300 and for columns it is considered as H/325 or H/250 depending upon load combination.

2.6.1 Lateral Sway

The lateral sway at the top of the building due to Wind loads should not exceed H/500, where H is the height of the building.

2.7 Fatigue Check

Fatigue phenomenon needs to be analysed only for those structural elements that are subjected to repetition of significant stress variation (under traffic load).

(i) **RCC and PSC structures -** As per clause 13.4 of IRS CBC.

(ii) **Steel Structures-**

a) In case of Metro live loads, as per clause 3.6 of IRS Steel Bridge /code shall govern. If λ* values are required to be used, the train closest to the actual train formation proposed to be run on the system shall be used. Otherwise, detailed counting of cycles shall be done.

b) For other cases as per Section 13 of IS:800.

*: Damage equivalence factors (As per IRS Steel Bridge Code)

2.8 Foundations

2.8.1 Types of Foundation

Considering the nature of ground, type of proposed structures, expected loads on foundations, the following foundations are to be considered.

a) Spread or pad footing
b) Raft foundation
c) Pile foundation

No matter the type of foundation to be adopted, the following performance criteria shall be satisfied:

1) foundation must not fail in shear

2) foundation must not settle by more than the settlements permitted as per Table-1 of IS:1904.

2.8.2 Design of Pile

IS 2911 (Part 1/Section 1)-2010 shall be followed for design of pile, load capacity etc.

Pile Settlement

Methods of estimating the settlement of deep foundations depend upon the type of deep foundation and the manner of transfer of loads from the structure to the soil. Theoretical estimation of settlement shall be done in accordance with IS:8009 (Part II) by integrating the vertical strain for the entire depth of soil and rock formation.

The settlement of each pile and/or pile group should be determined and it should be demonstrated that such total and/or differential settlement can be tolerated by the structure.

2.8.3 Foundations

IS: 1904 shall be followed for design of foundation in soil. The safe bearing capacity for the shallow foundations shall be calculated in accordance with IS:6403-1981.

Computation of Settlements of Foundations

The calculations for settlement of foundations shall be done as per:

- IS 8009 (Part I) for shallow foundations
- IS 8009 (Part II) for deep foundations

2.9 Design of Water Retaining Structure

- It should be designed as per IS IS 3370 (Part I), IS 3370 (Part II).

3. LIST OF DESIGN CODES AND STANDARDS

The designs of station buildings shall be carried out as per provisions of this Design Specifications. Reference shall be made to following codes for any additional information.

Order of preferences of codes shall be as follows: -

i. IS
ii. IRS
iii. IRC
iv. BS or Euro Code
v. AASHTO
vi. Other reference listed